CLINICAL · MOUTH & THROAT

Applied anatomy of the oral cavity

The mouth is a small room with many tenants: lip, cheek, gum, teeth, tongue, palate and floor, each with its own nerve, vessel and lymphatic route. Know where each subsite begins and ends, and where its lymph goes, and the examination of the mouth, the staging of its cancers and the planning of its neck dissections follow naturally.

AT A GLANCE

- The oral cavity ends at the circumvallate papillae, palate junction and anterior pillars.
- Tongue: lingual nerve and chorda tympani in front, glossopharyngeal behind.
- Lateral tongue may skip to level IV; floor of mouth crosses the midline.
- Referred otalgia comes through V3 and IX.

1 CHAPTER ONE Boundaries

Where the mouth ends and the pharynx begins.

1.1 Extent

The oral cavity runs from the vermilion border of the lips in front to the *oropharyngeal isthmus* behind: the junction of the hard and soft palate above, the line of the circumvallate papillae below, and the anterior tonsillar pillars and retromolar trigones at the sides. Everything behind this ring (soft palate, tonsils, base of tongue, valleculae) is oropharynx, a distinction that matters because the two regions are staged and treated differently.

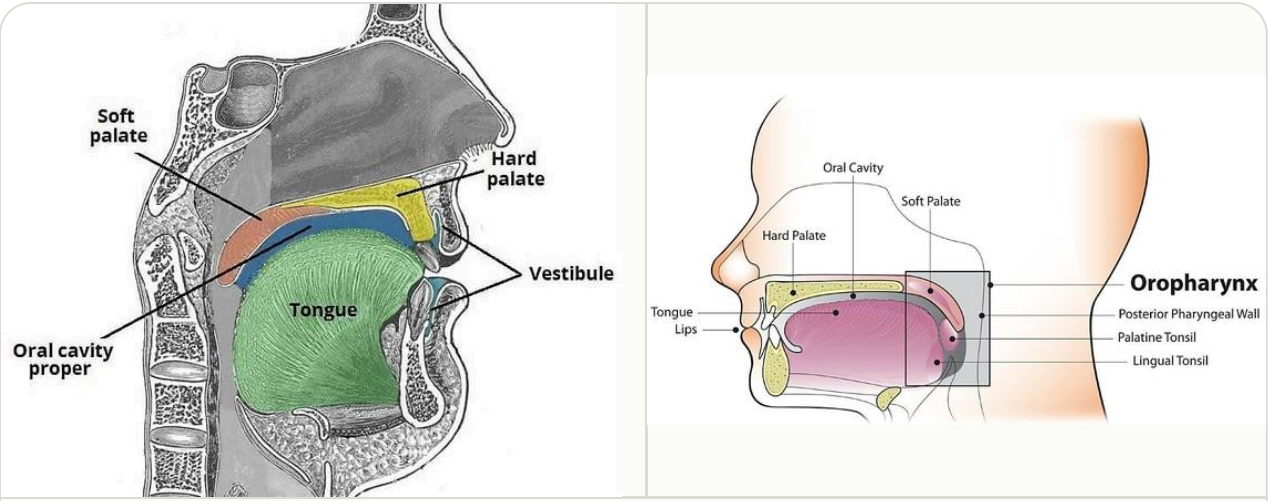


FIGURE 1 The vestibule and the oral cavity proper, and the boundary between the oral cavity and the oropharynx.

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1.2 Subsites

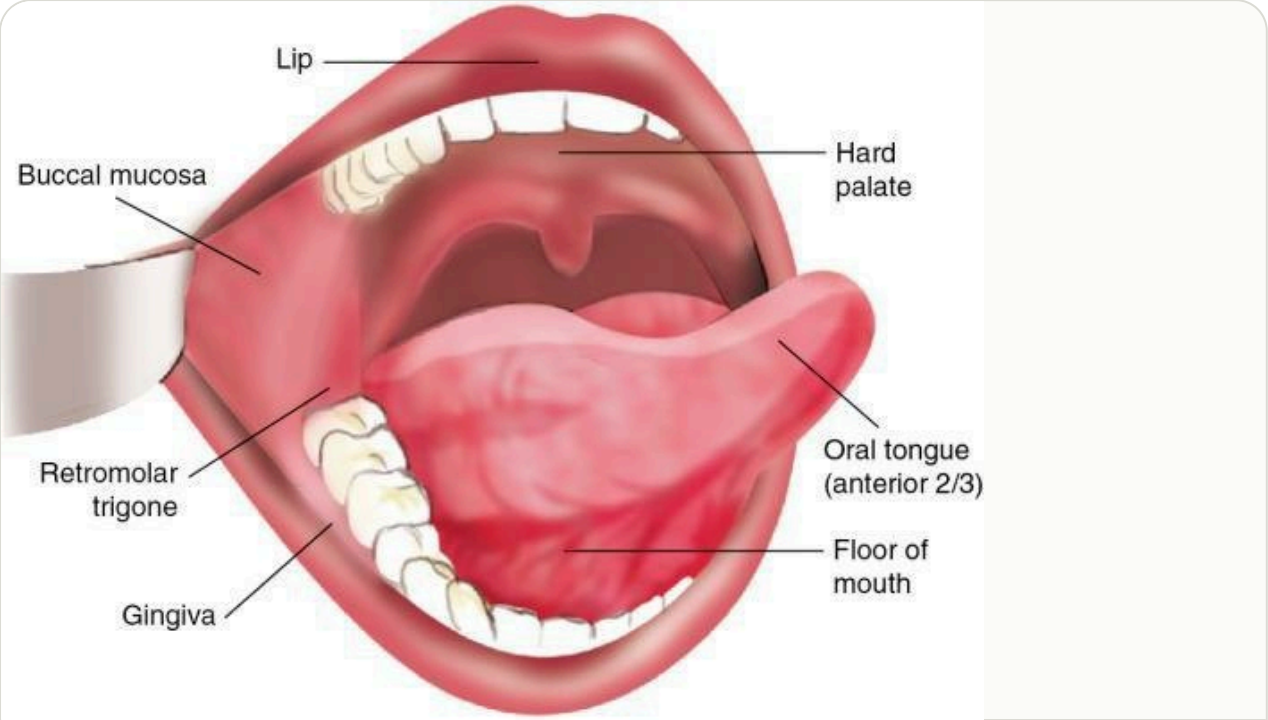


FIGURE 2 The subsites of the oral cavity: lip, buccal mucosa, gingiva, retromolar trigone, hard palate, oral tongue and floor of mouth.

Dhingra, fig. 42.1

- **Lips:** the mucosal lip, from the vermilion border to the line where the lips meet.
- **Buccal mucosa:** the lining of the cheeks and inner lips, back to the pterygomandibular raphe.

- **Upper and lower alveolar ridges (gingivae):** the mucosa over the tooth-bearing bone.
- **Retromolar trigone:** the triangle of mucosa over the front of the ascending ramus, its base behind the last molar and its apex at the maxillary tuberosity.
- **Hard palate:** the roof of the mouth.
- **Oral tongue:** the anterior two-thirds, in front of the circumvallate papillae, with its tip, lateral borders, dorsum and undersurface.
- **Floor of mouth:** the crescent between the lower gingiva and the undersurface of the tongue, with the frenulum and the sublingual papillae where the submandibular ducts open.

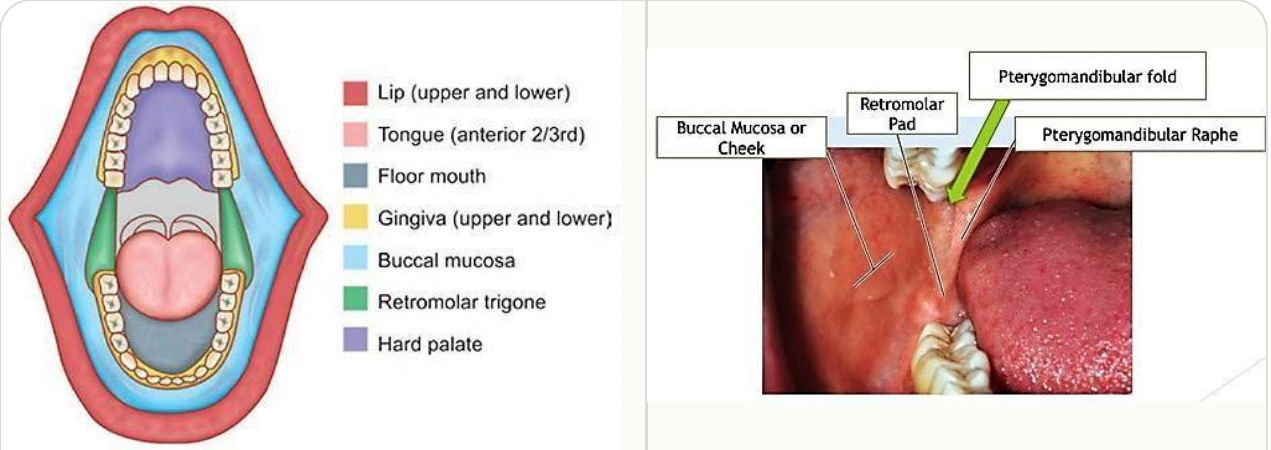


FIGURE 3 The subsites of the oral cavity, and the retromolar trigone with the pterygomandibular raphe.

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EXAMINING THE FLOOR OF MOUTH

The anterior floor is seen best when the patient lifts the tongue tip to the palate; the lateral floor when a tongue depressor pushes the tongue to the other side. Always palpate the floor bimanually: small cancers here are felt before they are seen.

1.3 Functions

Intake and chewing of food, the oral phase of swallowing, the shaping of speech, and protection against noxious substances through saliva, taste and the gag reflex.

2 CHAPTER TWO The lips

The gateway.

2.1 Surface anatomy

The *vermilion* is non-keratinised squamous epithelium without hair or sebaceous glands; its colour comes from the capillaries beneath. The vermilion border marks the change from skin to mucosa; the philtral ridges and Cupid's bow shape the upper lip; the two lips meet at the oral commissures.

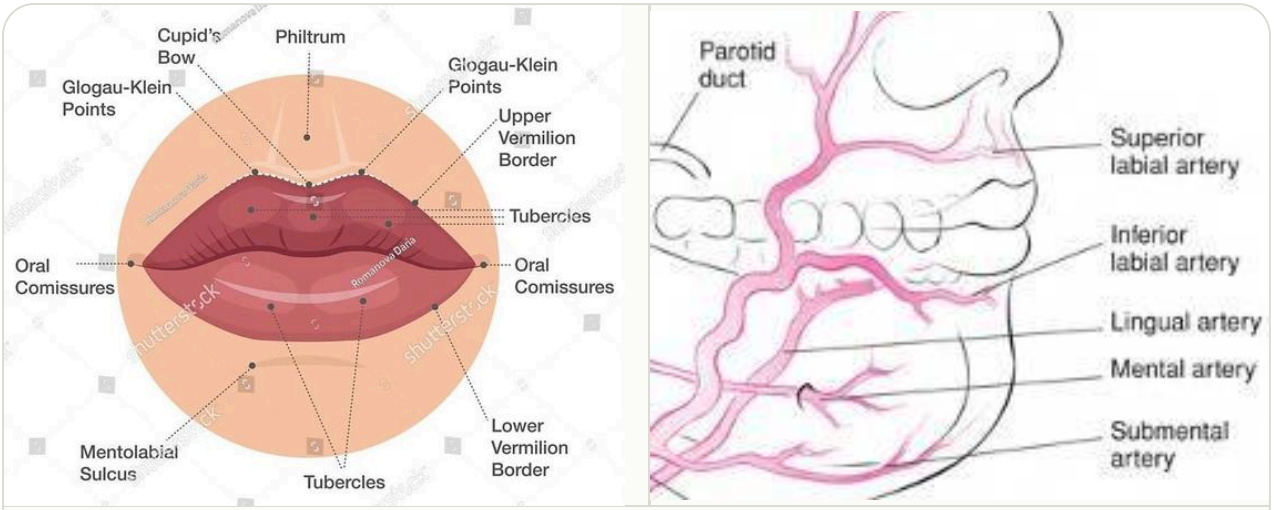


FIGURE 4 Lip landmarks, and the arterial supply of the lips from the superior and inferior labial arteries.

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2.2 Muscles

All are supplied by the facial nerve. The *orbicularis oris* forms the sphincter. At each commissure several muscles meet in a tendinous knot, the *modiolus*: orbicularis oris, buccinator, levator and depressor anguli oris, zygomaticus major and risorius. The lip elevators (levator labii superioris, levator labii superioris alaeque nasi, zygomaticus minor) and depressors (depressor labii inferioris, mentalis, platysma) complete the set.

2.3 Nerves, vessels and lymph

- **Sensation:** upper lip from the infraorbital nerve (V2), lower lip from the mental nerve (V3).
- **Arteries:** the superior and inferior labial branches of the facial artery.
- **Lymph:** the lower lip drains to the submandibular nodes, its midline also to the submental nodes, and it may cross the midline; the upper lip drains to the submandibular, preauricular and infraparotid nodes.

3 CHAPTER THREE Teeth, gums and jaws

What the dentist sees.

3.1 The tooth

- **Enamel:** the hardest outer layer, laid down by ameloblasts.
- **Dentine:** beneath it, formed throughout life by odontoblasts, and sensitive.
- **Pulp:** the inner core of vessels and nerves.
- **Cementum:** covers the root and anchors it through the periodontal ligament to the alveolar bone.

3.2 Numbers

Twenty deciduous teeth (no premolars) and thirty-two permanent teeth: in each quadrant two incisors, one canine, two premolars and three molars, the third molar being the wisdom tooth.

3.3 Gingiva

The attached gingiva is bound firmly to the alveolar bone; the looser alveolar mucosa lies beyond it.

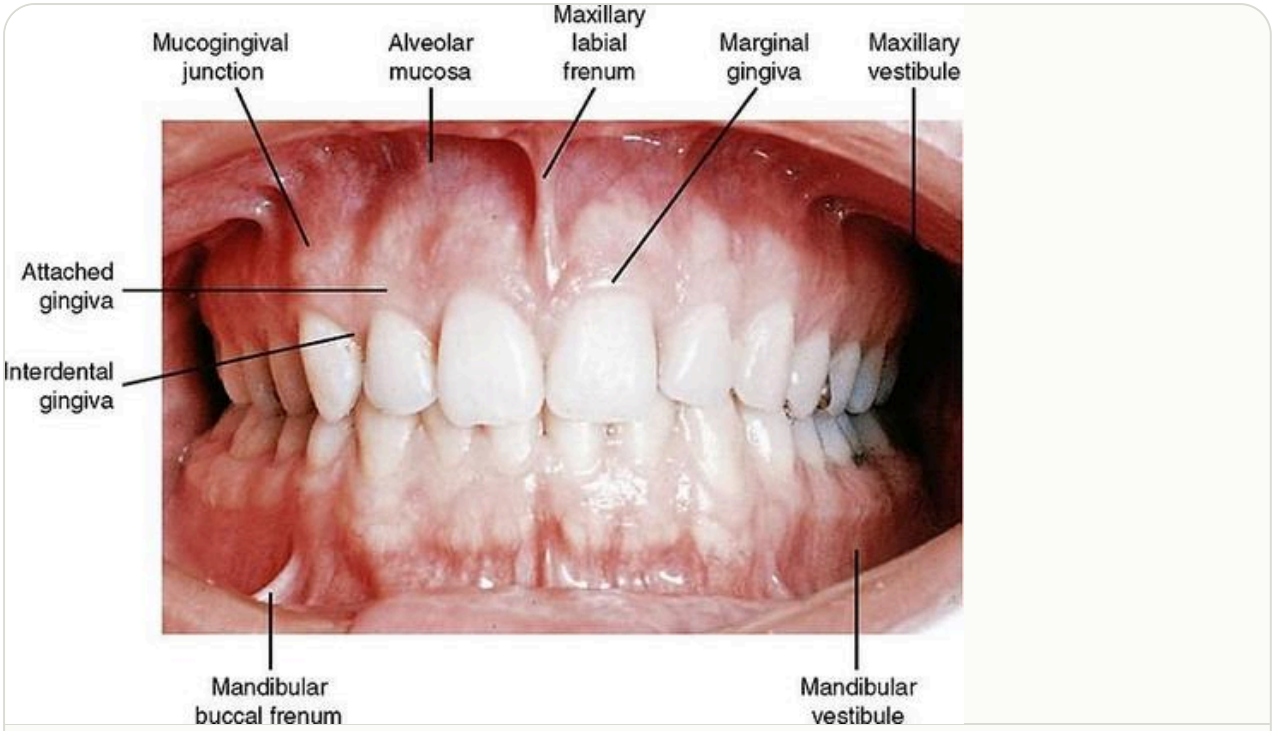


FIGURE 5 Gingiva and alveolar mucosa.
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3.4 Muscles of mastication

All supplied by the mandibular nerve (V3):

- **Masseter:** from the zygomatic arch to the angle and lateral ramus; elevates the mandible.
- **Temporalis:** from the temporal fossa to the coronoid process; elevates and retracts.
- **Lateral pterygoid:** from the greater wing of the sphenoid and lateral pterygoid plate to the condyle; opens the mouth, protrudes and moves the jaw from side to side.
- **Medial pterygoid:** from the medial side of the lateral pterygoid plate and the maxillary tuberosity to the medial angle; elevates and moves the jaw from side to side.

Infiltration of these muscles by infection or cancer produces *trismus*.

4 CHAPTER FOUR The tongue

The busiest muscle in the head.

4.1 Surface

The *sulcus terminalis* and the V of circumvallate papillae divide the oral tongue in front from the base of tongue behind. At the apex of the V lies the *foramen caecum*, from which the thyroid descends in the embryo; a lingual thyroid or thyroglossal cyst begins here. Behind lie the lingual tonsil and the valleculae, both oropharynx.

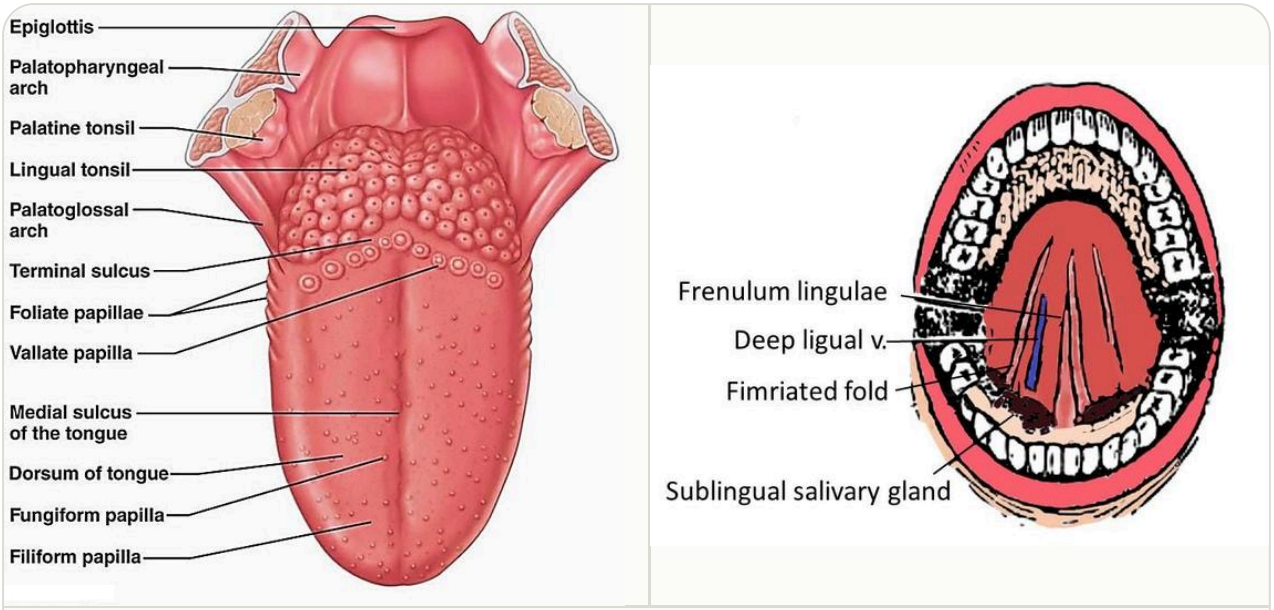


FIGURE 6 The dorsum of the tongue with its papillae, sulcus terminalis and lingual tonsil, and its undersurface with the floor of mouth.

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4.2 Papillae

- **Filiform:** the most numerous; texture, temperature and pain; no taste.
- **Fungiform:** mostly at the tip; taste buds on their upper surface.
- **Foliate:** on the lateral border; taste buds on their sides.
- **Circumvallate:** eight to twelve in a V in front of the sulcus terminalis; taste buds in their walls.

4.3 Muscles

The intrinsic muscles (longitudinal, transverse and vertical) change its shape; the extrinsic muscles move it.

EXTRINSIC MUSCLES OF THE TONGUE

MUSCLE	FROM	ACTION	NERVE
Genioglossus	Mental spine of the mandible	Protrudes and depresses; the bulk of the tongue	Hypoglossal
Hyoglossus	Hyoid bone	Depresses and retracts	Hypoglossal
Styloglossus	Styloid process	Retracts and elevates	Hypoglossal
Palatoglossus	Palatine aponeurosis	Raises the back of the tongue, closes the isthmus	Vagus (pharyngeal plexus)

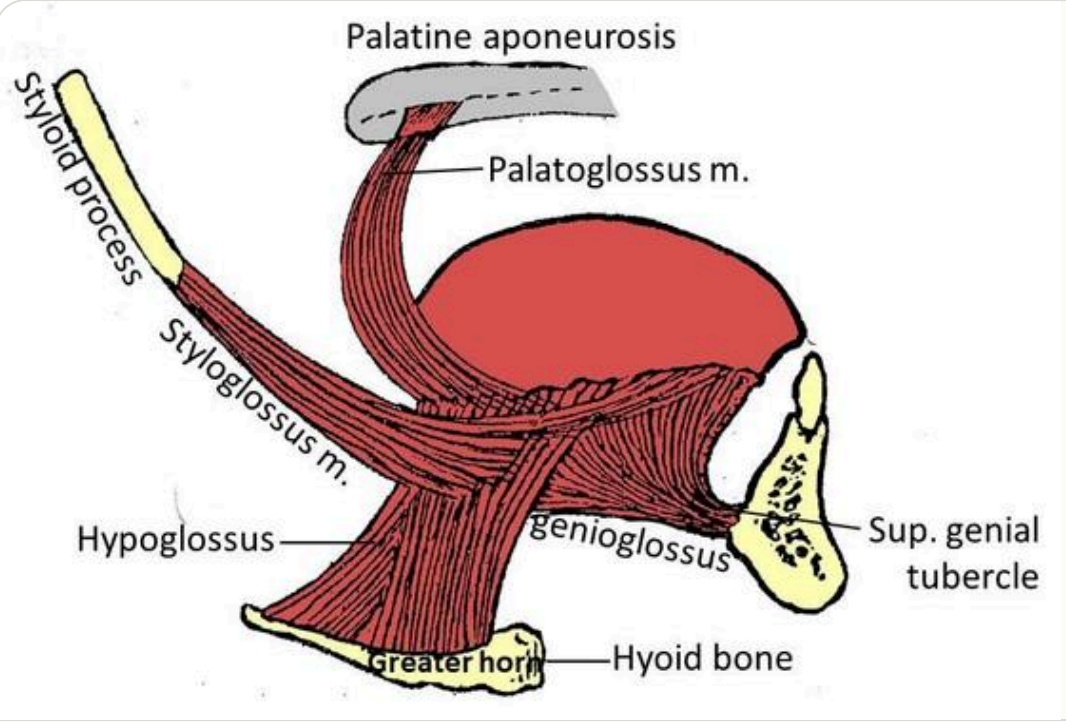


FIGURE 7 The extrinsic muscles of the tongue.
Anatomy atlas · Doss p. 179

TESTING THE HYPOGLOSSAL NERVE

With a lower motor neurone lesion the protruded tongue deviates towards the paralysed side, pushed across by the healthy genioglossus, and the paralysed half wastes and fasciculates.

4.4 Nerves and taste

- **Oral tongue:** touch, pain and temperature by the lingual nerve (V3); taste by the chorda tympani (VII) to the geniculate ganglion and nucleus solitarius.

- **Base of tongue:** both general sensation and taste by the glossopharyngeal nerve (IX).
- **Epiglottis and valleculae:** a little taste by the internal laryngeal branch of the vagus (X).

The five tastes are sweet, sour, salty, bitter and umami; flavour is taste combined with smell.

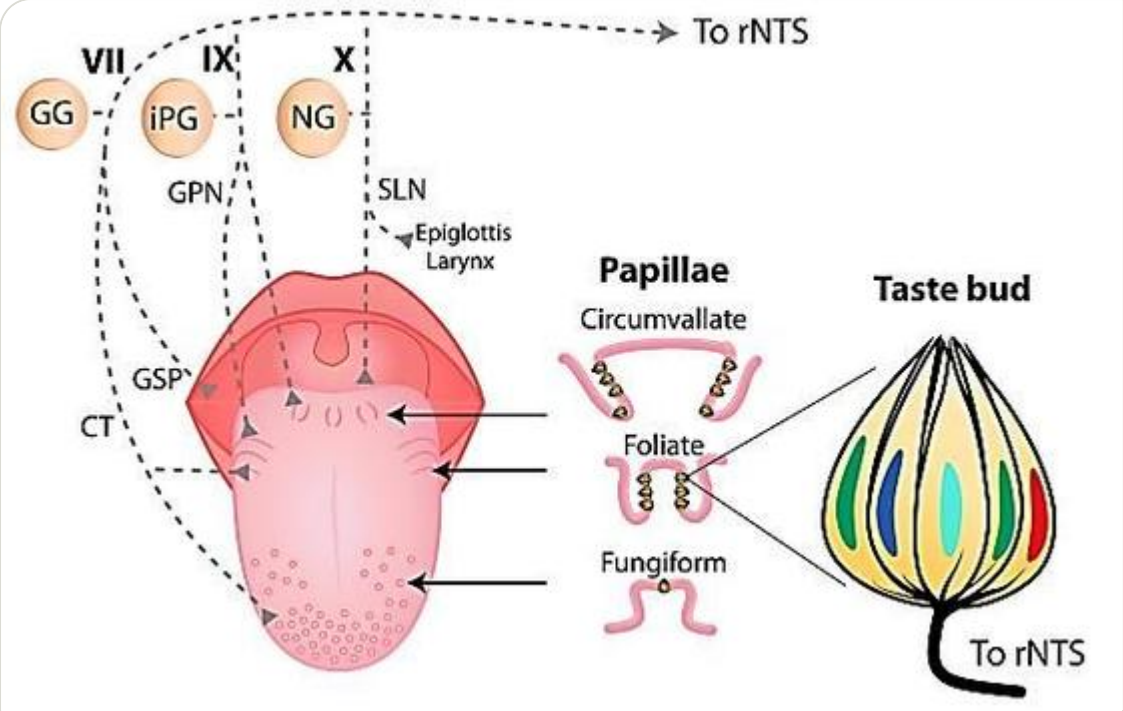


FIGURE 8 The nerve supply and taste pathways of the tongue.
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REFERRED OTALGIA

The lingual nerve and the auriculotemporal nerve are both branches of V3, and the glossopharyngeal nerve also supplies the middle ear through Jacobson's nerve. A cancer of the tongue or tonsil therefore often presents as earache with a normal ear.

4.5 Vessels and lymph

The lingual artery, second branch of the external carotid, supplies the tongue; the lingual vein and the vein accompanying the hypoglossal nerve drain it. Lymph from the tip goes to the submental and jugulo-omohyoid nodes, from the lateral border to the ipsilateral submandibular and deep cervical nodes (sometimes skipping

straight to level IV), and from the central tongue and base to the deep cervical nodes of both sides.

5 CHAPTER FIVE Palate and floor

Roof and floor.

5.1 Hard palate

The palatine processes of the maxillae in front and the horizontal plates of the palatine bones behind, covered by mucoperiosteum firmly bound to bone.

- **Incisive foramen:** in the midline behind the incisors; nasopalatine nerve and vessels.
- **Greater palatine foramen:** medial to the second or third molar; greater palatine nerve and vessels, the pedicle of palatal flaps.
- **Lesser palatine foramina:** behind it, to the soft palate.

Sensation is from the nasopalatine and greater palatine nerves (V2); blood from the greater palatine artery; lymph to the upper deep cervical and lateral retropharyngeal nodes, the front of the palate also to the submandibular nodes.

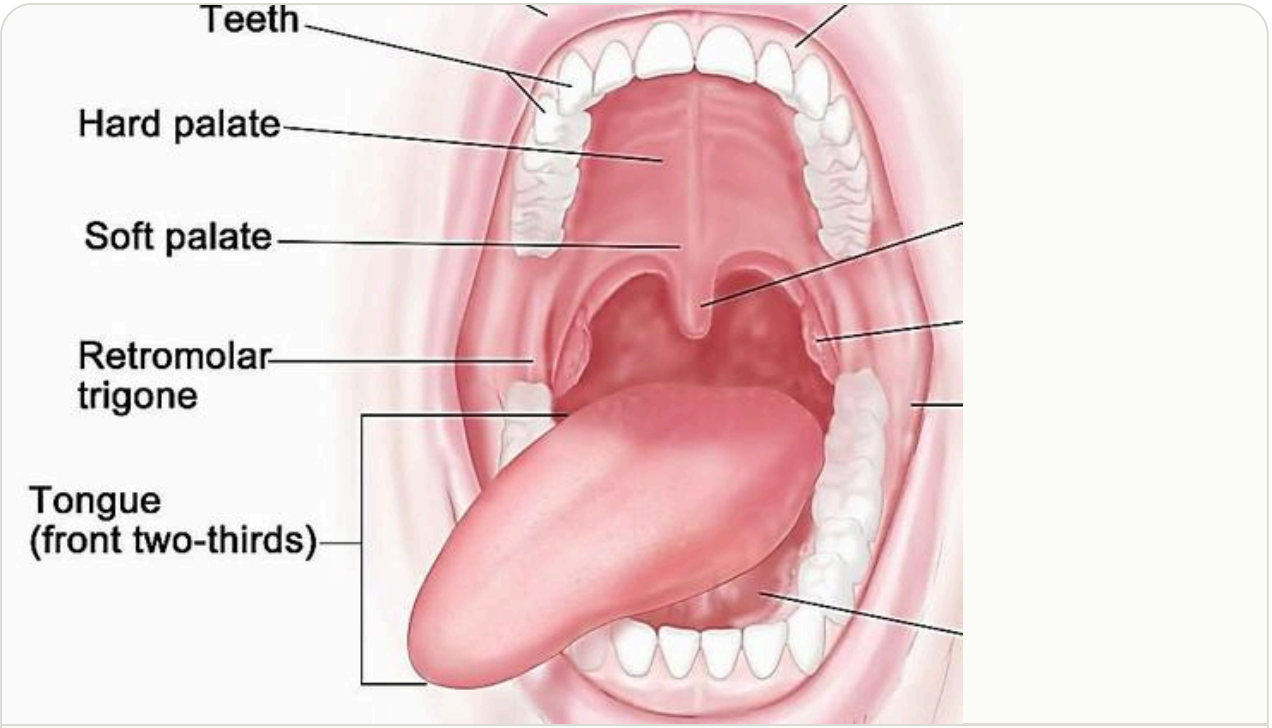


FIGURE 9 The hard palate, soft palate, retromolar trigone and oral tongue.
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5.2 Floor of mouth

The mylohyoid muscle forms the diaphragm of the floor. Above it lie the sublingual glands, the submandibular duct, the lingual nerve looping beneath the duct, and the hypoglossal nerve; below it, the submandibular space. The anterior floor drains to the submandibular nodes and crosses the midline freely; the posterior floor drains to the upper deep cervical nodes.

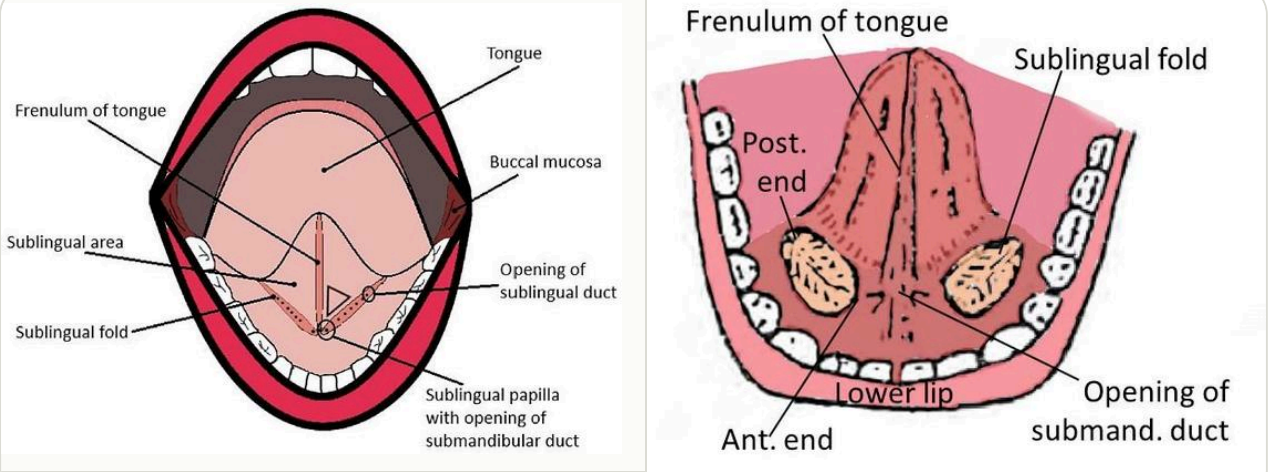


FIGURE 10 The floor of the mouth with the sublingual papillae and folds, and the openings of the ducts.

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5.3 Saliva

One to one and a half litres a day. At rest about two-thirds comes from the submandibular glands; when stimulated, more than half comes from the parotids. The parotid (Stensen) duct opens opposite the upper second molar; the submandibular (Wharton) duct beside the frenulum; the sublingual glands by many small ducts into the floor. Minor salivary glands line the lips, cheeks, palate and tongue, and are the source of the palate's salivary tumours.

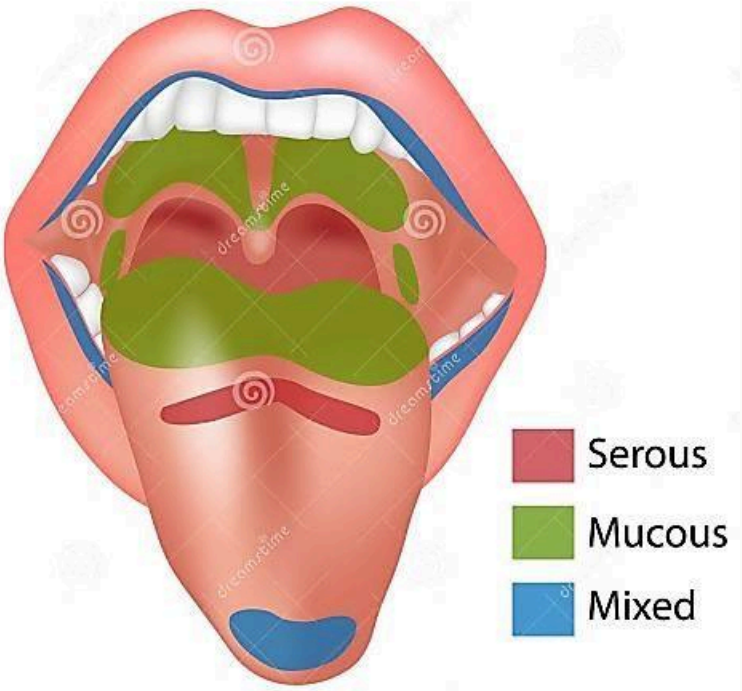


FIGURE 11 Distribution of serous and mucous minor salivary glands.
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6 CHAPTER SIX Lymphatic drainage

The map for the neck.

LYMPHATIC DRAINAGE OF THE ORAL CAVITY

SUBSITE	FIRST NODES
Lower lip	Submental (midline) and submandibular (lateral); may cross the midline
Upper lip	Submandibular, preauricular and infraparotid
Buccal mucosa	Submental and submandibular
Alveolar ridges	Submental and submandibular
Hard palate	Upper deep cervical and lateral retropharyngeal; front of palate to submandibular
Floor of mouth	Anterior to submandibular, crossing the midline; posterior to upper deep cervical

SUBSITE	FIRST NODES
Tongue tip	Submental and jugulo-omohyoid
Lateral tongue	Ipsilateral submandibular and deep cervical, sometimes skipping to level IV
Central tongue and base	Deep cervical nodes of both sides

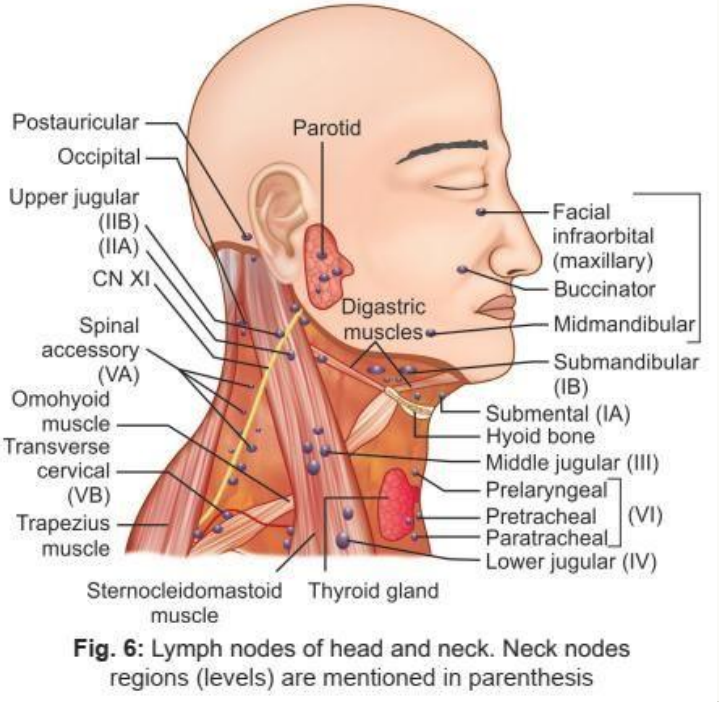


Fig. 6: Lymph nodes of head and neck. Neck nodes regions (levels) are mentioned in parenthesis

FIGURE 12 The lymph node levels of the neck.

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7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Boundaries** of the oral cavity and its subsites.
- **Nerve supply of the tongue:** general sensation and taste, front and back.
- **Extrinsic muscles** and the test of the hypoglossal nerve.

- **Lymphatic drainage** of the lip, tongue and floor of mouth.
- **Referred otalgia** from the mouth and pharynx.

7.2 Common pitfalls

COMMON PITFALLS

- Calling the base of tongue or the tonsil "oral cavity".
- Forgetting that the lateral tongue can skip to level IV nodes.
- Inspecting the floor of mouth without palpating it.

Written for this app from Oxford Specialist Handbook ch. 6, Dhingra ch. 42 and K. J. Lee ch. 36. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

ORAL CAVITY & TONGUE · AFTER SURGERY

Squamous cell carcinoma: oral cavity, oropharynx and larynx

IMAGING FINDINGS

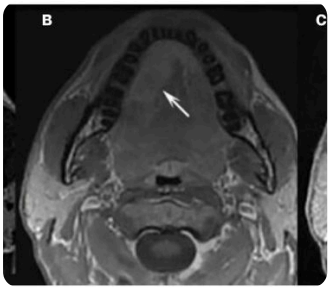
- Enhancing mucosal mass; depth of invasion in the tongue
- Crossing the lingual septum; neurovascular bundles
- Mandible cortex (CT) and marrow (MRI)

TOP DIFFERENTIALS

- Lingual tonsil hyperplasia
- Abscess
- Minor salivary gland tumour

CHECKLIST

- MRI for oral and oropharyngeal tumours
- Nodes on both sides; chest CT



Axial MRI of an oral cavity SCC shows a solid mass involving the left lateral border of the tongue showing intermediate signal intensity on T2…



Coronal T1WI MR in a woman with a painful right tongue ulcer shows subtle low signal intensity at the lateral tongue margin. There is no evidence …

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 504.